

Roudy Hajo

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Final year Computer Science student at Polytech Nantes School of Engineering with a passion for software development, network and cloud computing. Experienced in developing full-stack web applications, REST APIs, and conducting research in Database Systems. Strong problem-solving skills and a proven track record of delivering high-quality projects.

EDUCATION

Polytech Nantes School of Engineering <i>Master's Degree in Computer Science specialization in Network, Security, Cloud Computing</i>	Nantes, France Sep. 2024 – Aug 2027
University Institute of Technology of Nantes <i>Technical Associate's Degree in Computer Science</i>	Nantes, France Sep. 2022 – Jun 2024

EXPERIENCE

Research Intern in Database Systems <i>Coimbra Institute of Engineering</i>	May 2026 – Aug 2026 Coimbra, Portugal
- Designed a decentralized multi-agent architecture utilizing Quantum Neural Networks (QNN) and reinforcement learning to optimize database index tuning. - Engineered an adaptive centralized learning router to eliminate workload skew and dynamically balance queries across replicas.	
Web Developer Intern <i>Nantes University – Digital and Information Systems Department (DSIN)</i>	May 2025 – July 2025 Nantes, France
Updated and adapted the Symfony web application 'Syllabus' to dynamically present the university's course offerings to students.	
Web Developer Intern <i>Kookline – Web Development Agency</i>	Apr 2024 – June 2024 Nantes, France
Developed a responsive website using the WordPress Content Management System (CMS) for skydiving reservations and online services.	
Academic Tutor <i>University Institute of Technology of Nantes</i>	Oct 2023 – July 2024 Nantes, France
Mentored and guided first-year students in fundamental computer science courses, focusing on algorithms, core programming principles, and mathematics.	

PROJECTS

TODO App <i>Spring Boot, Java, Maven, Angular</i>	Oct 2025 – June 2026
- Developed a responsive full-stack task management web application supporting secure CRUD operations. - Implemented dynamic client-side routing with Angular and engineered reactive RESTful APIs with Spring Boot.	
SketchOscent <i>HTML/CSS, JavaScript, jQuery, Webpack, Python, RDKit, PubChem</i>	Sep 2025 – May 2026
- Enhanced a cheminformatics web application to model chemical odors as graphs and visualize molecular data. - Calculated and mapped chemical functional groups to specific scent profiles using Python, RDKit, and PubChem. - Integrated an automated code obfuscation pipeline into Webpack to effectively secure intellectual property.	
Dice Wars <i>C++</i>	Nov 2025 – Jan 2026
Engineered a turn-based strategy game featuring heuristic-driven AI opponents and a procedural map generation system.	
PolyRisk <i>Python</i>	Oct 2025 – Nov 2025
Developed intelligent AI agents for the Risk board game utilizing custom heuristics and Monte Carlo Tree Search (MCTS).	
Football App <i>HTML/CSS, JavaScript, Node.js, Express.js, Passport.js, React, Tailwind CSS</i>	Mar 2024 – Apr 2024
- Built a full-stack platform fetching and caching external REST API football match results and live statistics. - Engineered a secure multi-user betting competition game utilizing Passport.js for session authentication.	
Botanicia <i>HTML/CSS, JavaScript, PHP, Symfony, MariaDB, Apache</i>	Nov 2023 – Jan 2024
Developed a full-stack e-commerce web application for gardening products with CRUD features and user authentication.	

TECHNICAL SKILLS

Languages: Java, Kotlin, Python, C/C++, SQL (Oracle, MariaDB), MongoDB, JavaScript, HTML/CSS, R
Frameworks: Spring Boot, Angular, React, Node.js, Flask, JUnit, WordPress, FastAPI
Developer Tools: Git, Docker, Kubernetes, Microsoft Azure, VS Code, Visual Studio, IntelliJ,